Practice Problems

Question-1

Let A be a real symmetric matrix. Prove that eigenvectors corresponding to distinct eigenvalues are orthogonal. Then show that A admits an orthonormal basis of eigenvectors.

Question-2

For

$$A = \begin{pmatrix} 4 & 2 \\ 1 & 3 \end{pmatrix}$$

1. Find eigenvalues.
2. Determine if the matrix is diagonalizable.
3. Compute A^5 using diagonalization.

Question-3

Let A be symmetric. Prove the equivalence:

$$\text{all eigenvalues of } A \text{ are positive} \Leftrightarrow x^T A x > 0 \text{ for all non-zero } x.$$

(Hint: Solve the Bonus Problems first)

Question-4

Prove that a square matrix A is invertible if and only if 0 is not an eigenvalue of A. Use this to show that $\det(A)$ equals the product of eigenvalues.

Question-5

Show that $\text{trace}(P^{-1}AP) = \text{trace}(A)$ for any invertible matrix P. Deduce that the trace of a matrix equals the sum of its eigenvalues, even if the matrix is not diagonalizable.

Question-6

Let λ be an eigenvalue of A with eigenvector v. Prove that for any polynomial $p(x)$,

$$p(A)v = p(\lambda)v.$$

Use this to prove the Cayley–Hamilton theorem for diagonalizable matrices.

Question-7

Find the eigenvalues and eigenvectors of

$$A = \begin{pmatrix} 3 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 3 \end{pmatrix}$$

(Hint: Try the vector $(1, 1, 1)$)

Question-8

Prove that the rank of a symmetric matrix equals the number of non-zero eigenvalues (counting multiplicity). Explain why this statement fails for general matrices.

Question-9*

Let A be symmetric positive definite and consider

$$f(x) = \frac{1}{2}x^T Ax - b^T x.$$

1. Prove that $f(x)$ has a unique global minimizer.

2. Show that the minimizer satisfies $Ax = b$.
3. If $\lambda_{\min}$ and $\lambda_{\max}$ are the smallest and largest eigenvalues of A, prove:

$$\lambda_{\min}\|x\|^2 \leq x^T Ax \leq \lambda_{\max}\|x\|^2.$$

4. Explain why the ratio $\lambda_{\max}/\lambda_{\min}$ determines the convergence speed of gradient descent.

"A matrix and its transpose agree more than group project partners."

Bonus Problems

Let A be an $n \times n$ real matrix. Prove or disprove the following properties involving a matrix and its transpose:

1. $\text{trace}(A) = \text{trace}(A^T)$
2. $\det(A) = \det(A^T)$
3. $\text{rank}(A) = \text{rank}(A^T)$
4. A and A^T have the same eigenvalues
5. $A^T A$ is invertible if and only if the columns of A are linearly independent
6. $\ker(A^T A) = \ker(A)$
7. For any vector x, $\|Ax\|^2 = x^T A^T Ax$
8. AA^T is also symmetric positive semidefinite
9. All eigenvalues of $A^T A$ are non-negative
10. $A^T A$ and AA^T have the same non-zero eigenvalues